

Questions with Answer Keys

MathonGo

Q1 (25 July 2021 Shift 1)

In the Young's double slit experiment, the distance between the slits varies in time as

$d(t) = d_0 + a_0 \sin \omega t$; where d_0 , ω and a_0 are

constants. The difference between the largest fringe width and the smallest fringe width obtained

over time is given as :

(1) $\frac{2\lambda D(d_0)}{(d_0^2 - a_0^2)}$

(2) $\frac{2\lambda D a_0}{(d_0^2 - a_0^2)}$

(3) $\frac{\lambda D}{d_0^2} a_0$

(4) $\frac{\lambda D}{d_0 + a_0}$

Q2 (27 July 2021 Shift 1)

In Young's double slit experiment, if the source of light changes from orange to blue then:

(1) the central bright fringe will become a dark fringe.

(2) the distance between consecutive fringes will decrease.

(3) the distance between consecutive fringes will increase.

(4) the intensity of the minima will increase.

Q3 (27 July 2021 Shift 2)

The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is mm. [Refractive index of air = 1.0003, wavelength of yellow light in vacuum = $6000\text{\AA}$. . .]

Answer Key

Q1 (2)

Q2 (2)

Q3 (2)

Hints and Solutions

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Q1

Fringe Width, $\beta = \frac{\lambda D}{d}$ $\beta_{\text{aax}} \Rightarrow d_{\text{min}}$ and $\beta_{\text{min}} \Rightarrow d_{\text{max}}$ $d = d_0 + a_0 \sin \omega t$ $d_{\text{max}} = d_0 + a_0$ and $d_{\text{min}} = d_0 - a_0$ $\beta_{\text{min}} = \frac{\lambda D}{d_0 + a_0}$ and $\therefore \beta_{\text{max}} = \frac{\lambda D}{d_0 - a_0}$

Q2

Fringe width = $\lambda D/d$ as λ decreases, fringe width also decreases

Q3

Thickness $t = n\lambda$ So, $n \lambda_{\text{vac}} = (n+1) \lambda_{\text{air}}$ $n \lambda = (n+1) \frac{\lambda}{\mu_{\text{air}}}$ $n = \frac{1}{\mu_{\text{air}} - 1} = \frac{10^4}{3}$ $t = n\lambda$ $= \frac{10^4}{3} \times 6000 \text{Å} \dots$ $= 2 \text{ mm}$